

Peiyuan Qi

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PROFESSIONAL EXPERIENCE

NVIDIA Corporation (NVIDIA)

Santa Clara, CA

Senior Engineer, System Software – Hardware Infra - Machine Learning.

Feb. 2025 – Present

- Designed and agentially implemented the review infra end to end for agentic Reinforcement Learning data review.
- Designed and implemented the modern agent which serves as Chief of Staff of NVIDIA with more than 50k core python backend code features within 20 days with help of Claude Code and Codex, with proper design of layered backend, **concurrency** management, easy installation, full suite of GUI with **React/Tailwind/vsix**, with proper set of unit/regression tests.
- Led the ML study sessions in the Org to explore the techniques in **Attention Residual**, **mHC**, and **LoopLM**, etc.
- Developed [the OneLogger v2](#) with the team, a python library to monitor training job **GPU utilization** efficiency.
- Collaborated with internal customers like Cosmos, NeMo to onboard the OneLogger, to achieve 60+% coverage.
- Led and grow HWINF Applied AI team from 2 to 12 members in 2 months to facilitate AI transformation.
- Effectively using LLM with monthly API cost scale at 20k+ USD level.

ByteDance Inc. (TikTok)

San Jose, CA

Senior Backend Software Engineer – Global E-Commerce – Warehouse.

Oct. 2024 – Feb 2025

- Led the warehouse management system outbound team (RocketMQ, FaaS, micro services with Peak QPS 3k+).
- Migrated the SQL databases and HDFS due to regulation within 1 week, while maintained the high availability.

Google LLC. (Google)

Sunnyvale, CA

Software Engineer – Google Cloud - ML, Systems, Cloud AI (MSCA).

Jun. 2021 – Oct 2024

- **Managed** multiple software programs as program Tech Lead of the domain services in the team, for data center modeling.
- **Launched** multiple successful cross-organizational programs **without outages**, which outcomes are featured in external documents and news: [GKE Compact Placement](#), [TPU v5e](#), [A3 GPU](#), [Trusted Partner Cloud program](#), saving the company avg \$XX million per projects, and opening market of \$X billions, cooperated closely with principle engineers. Reacted quickly with simultaneous business demands for New Product Introductions and pushed multiple projects to progress at the same time.
- Led the **key transformations** of Google's GPU Network deployment strategy to better meet the hardware needs of [NCCL](#) for Google TCPDirect, the plugin.
- Defined the **API** following Google [AIP](#) and **SLO** in the perspectives of availability and latency for backend services.
- Acted as the go-to person in team for knowledge for Google infra tools, in term of **ACL**, **metrics**, **monitoring**, **alerting**, **CICD**.
- Crafted the math solver (**CP-SAT/MIPS**) modeling problem based on constraints in for Power, Network, Space, Cooling.
- **Led** new team members since 2022. Projects include using internal **LLM API**, **Google OneTwo** to build RPC for non-tech PM to use team services, building full-stack web features. Also explored the possibility to deploy **Pytorch** in Google infra.
- Explored the possibility of using the **Embedding** models for data center planning entities (orders, forecast data, machine shapes, rack physical data, network demands, etc) representations. Solved real deployment problems using algorithms like **Hungarian**.
- Transformed the existing DOM object rendering method to WebGL with **Entity Component System** model using **threejs** as a team, with **Data Oriented Design** principle. Developed frontend features with **Angular** and **Typescript**, backend features with Python and Java servers. Provided monitoring, metrics reporting and alerting supports with internal tools.

Electronic Arts Inc. (EA)

Redwood City, CA

Software Engineer – EA Data & AI Platform

Feb 2020 – Jun 2021

Software Engineering Internship

May 2019 – Aug 2019

- Developed the container images for legacy Spring Boot socket server and migrated the infra to **AWS** to save more than 50% average resource utilization. And conducted the **stress test** up to 5k QPS to ensure the robustness and the autoscaling ability.
- Built the monitor, reporting, alerting with the KairosDB, CloudWatch, **Elasticsearch**, Kibana.
- Owned the **Kafka** and **Apache Storm** for the real-time metrics processing supporting the billing of the games.

Interned at NVIDIA Shanghai and Intel Shanghai

Aug 2017 – Aug 2018

- Worked on RESTful API design and command line tool delivery in open-source community.
- Applied doc2vec and WMD algorithms to perform the Bug Triage tasks to improve team debugging efficiency.

OTHER PROJECTS

Phytophraphy, a Cosmos reason 2 driven photo app for better photography tips.

Feb 2026

EDUCATION

University of Southern California (USC), Los Angeles, CA

Master of Science in Computer Science

Aug 2018 – Jan 2020

Shanghai Jiao Tong University (SJTU), Shanghai, China

Bachelor of Science in Electrical and Computer Engineering Graduate with Honors

Sept 2014 – Aug 2018

University of Michigan – Shanghai Jiao Tong University Joint Institute